Arun Sekhar

Bachelor of Engineering in Electronics & Communication Engineering
Amrita College of Engineering, Erachakulam

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Professional Summary

Dedicated electronics and communication engineer with hands-on experience in computer hardware maintenance, embedded systems, software and hardware interfacing, and Arduino programming. Looking for full-time positions in embedded systems, semiconductors, cybersecurity, hardware validation, or robotics.

Education

Year	Degree/Certificate	Institute	CGPA/Percentage
2023	HSC - Computer Science	Government Higher Secondary School, Kalkulam	76.66%
2023-2027	B.E. Electronics and Communication Engineering	Amrita College of Engineering, Erachakulam	7.95/10.00

Experience

- **Self-Employed**
  Part-Time Computer Technician Remote/Local
  – Repaired, assembled and upgraded desktop computers and laptops.
  – Provided Bug Fixing and technical support.

Projects

- **HealthCost AI – AI-Powered Healthcare Cost Analysis Platform**
  React, Vite, Node.js, MongoDB
  – Developed a full-stack healthcare web application to help patients and families analyze and understand medical bills.
  – Implemented medical bill uploads with support for PDF,docs and image formats such as jpg and png.
  – Developed bill data extraction from images and calculation validation to identify potential arithmetic discrepancies using Gemini API.
  – Integrated AI-powered explanations to simplify complex medical charges for patients.
- **Smart Door Lock System with App Integration**
  Arduino UNO, HC-05 BT module,Servo
  – Designed and developed a smart door locking mechanism utilizing an Arduino UNO microcontroller and an arduino mini servo motor for physical actuation.
  – Integrated an HC-05 Bluetooth module to enable seamless wireless lock and unlock functionality via a dedicated mobile application.(Lock Reaction Time 0.86 seconds)
  – Incorporated multi-factor authentication by implementing both fingerprint and PIN verification methods for enhanced security.
- **Retro FM – Android FM Radio Application**
  Reverse Engineering & Native Interface
  – Developing an Android FM radio application for MediaTek-based devices using reverse engineering and native C++ (JNI).
  – Analyzed proprietary FM radio libraries (.so) using Ghidra to understand hardware communication and driver interactions.
  – Integrated Android Studio, Kotlin, CMake and JNI to interface with native FM libraries and device nodes.
  – Investigated Linux device drivers, Android HAL, and low-level hardware access for FM tuner functionality.
  – Debugged native code, build configuration, and Android system-level integration using Logcat and GDB.

Technical Skills

- **Programming Languages**: C (Basic), C++ (Basic), Python (Basic)
- **Embedded**: Arduino, Embedded C, Microcontrollers
- **Hardware**: PC Assembly, Hardware Diagnostics, BIOS Configuration, Circuit Debugging
- **Tools**: Arduino IDE, MATLAB, Git (Basic), MS Office
- **Operating Systems**: Windows, Linux (Basic)
- **Networking**: Basic Configuration and Troubleshooting

Relevant Coursework

- **Core Subjects**: Embedded Systems, Digital Electronics, C Programming, Python Programming
- **Cybersecurity and Ethical Hacking**:

Internship Experience

- **Shristi Innovative** *1 Month*
  Embedded Systems Design Intern

Interests

Embedded Systems, Semiconductor Engineering, Automation, Software Development, Cybersecurity, CCNA